INCH-POUND

MIL-DTL-43049E(IH)
20 December 2023
SUPERSEDING
MIL-B-43049D(IH)
20 November 1992

DETAIL SPECIFICATION

BACKING PLATE, INSIGNIA, BRANCH OF SERVICE AND CAP, INFANTRY, ARMY

This specification is approved for use by all
Department and Agencies of the Department of Defense

1: SCOPE

1.1 : Scope this specification covers the plastic backing plates used by Army Infantry enlisted personnel (see 6.1).

1.2 : Classification Backing plates will be of the following types(see 6.2).

Type I - Backing plate, insignia, service cap, infantry Blue.
Type II – Backing plate, insignia, Branch of service collar, infantry Blue.

2: APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

NN-P-71 - Pallets, Material Handling Wood Stringer Construction 2-Way and 4-Way (Partial)
MIL-DTL-15011 - Pallets Material Handling, Wood Post Construction, 4-Way Entry

Comments, suggestions, or questions on this document should be addressed to The Institute of Heraldry, ATTN: Technical and Production Division, 9325 Gunston Road, Room S-108, Fort Belvoir, VA 22060-5579 or email to usarmy.belvoir.hqda.mil.tioh-webmaster@mail.mil. You may verify the currency of this specification using the ASSIST Online database at https://assist.dla.mil.

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FEDERAL STANDARDS

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD- 147 – Palletized Unit Load

SPECIFICATION SHEETS
(See Supplement 1 for list of specification sheets.)

(Copies of these documents are available online at https://quicksearch.dla.mil).

2.1.2 : Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

THE INSTITUTE OF HERALDRY DRAWINGS

B-2-51 – Backing Plate, Insignia, Cap & Collar, U.S. Army Infantry

(Copies of standards, drawings and Chip sets are available from The Institute of Heraldry, ATTN: Technical and Production Division, 9325 Gunston Road, Room S-108, Fort Belvoir, VA 22060-5579 or usarmy.belvoir.hqda.mi.tioh-webmaster@mail.mil.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY CONTROL

ANSI/ASQC Z1.4 1993 - Sampling Procedures and Tables for Inspection by Attributes

(Copies for this document are available online at https://acq.org)

ASTM INTERNATIONAL

ASTM-D-570 - Plastic, Water Absorption of
ASTM-D-3951 – Standard Practice for Commercial Packaging

(Copies for these documents are available online at https://www.astm.org)

AMERICAN AND EFIRD, LLC
DoD Standard Shades for Heraldic Yarns, 2012 Edition

(Copies for this document are available online at www.salesRA@amefird.com)

SAE INTERNATIONAL
SAE-AMS-STD-595 - Colors Used in Government Procurement

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(Copies for this document are available online at https://sae.org)

2.3: Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

3: REQUIREMENTS

3.1 : First article When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 : Standard sample Unless otherwise specified (see 6.2), the contracting officer shall furnish a standard sample of the backing plate which shall be used for matching color and finish only. Variation from this specification may appear in the standard sample; however, in such cases the specification shall govern.

3.3 : Materials The Materials shall conform to the referenced specifications and the requirements specified herein.

3.3.1 : Plastic The backing plate shall be molded from virgin plastic material. Cellulose Nitrate or Cellulose Acetate Butyrate shall not be used.

3.4 : Construction The backing plate, type I or type II, shall be molded in accordance with the details and dimensions of TIOH Drawing B-2-51 (see Figure 1). The backing plates shall meet the performance requirements specified herein.

3.4.1 : Stability When tested as specified in 4.7.1, the backing plate shall not twist, chip, crack, break, or blister and shall be free from warp. When the backing plate is placed on a surface plate with the beveled edge upward, at least 75 percent of the perimeter of the backing plate shall be in contact with the plate and there shall be no gap exceeding 1/32 inch.

3.4.2 : Water absorption When tested as specified in 4.7.2, the water absorption shall not be greater than 2.5 percent.

3.4.3 : Color The color of type I or type II backing plates shall be light blue approximating Chip No. 15177 from SAE-AMS-STD-595 or the standard sample (see 3.2).

3.4.4 : Knockout pin mark Knockout pin marks and breakoff marks shall be trimmed flush within 1/64 inch of adjacent areas. There shall be no sharp edge. The marks shall not be located in areas which are visible under actual use.

3.5 : Finish The backing plates shall be free of warp, twist, cracks, blisters, scratches, sink mark, flash, fins and roughness at breakoff point and shall met the quality provisions as specified herein. The occurrence of defects shall not exceed the applicable acceptable quality levels.

3.6 : Workmanship the backing plates shall be free of warp, twist, cracks, blisters, scratches, sink marks, flash fins and roughness at breakoff point and shall meet the acceptable quality provisions established by this specification.

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4: VERIFICATION

4.1 : Inspection responsibility. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 : Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, not does it commit the Government to accept defective material.

4.2 : Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 : Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.5 and 4.6.

4.4 : First article sample inspection. Inspection and testing of the first article sample (see 3.2), shall be made of a completely finished backing plate for all provisions of this specification applicable to the end product examination and tests.

4.5 : Quality conformance inspection. Inspection shall be in accordance with the 'provisions set forth in MIL-STD-105, except where otherwise indicated (see 6.2).

4.5.1 : Inspection of components.

4.5.1.1 : Certification of compliance. The plastic material may be accepted based on the contractor's certificate of compliance that the plastic material meets the requirements of 3.3.1.

4.5.2 : Inspection of the end item.

4.5.2.1 : Visual examination of the backing plate. Examination shall be made at approximately 16 to 22 inches. The defects found during examination shall be classified in accordance with 4.5.2.1.1 and 4.5.2.1.2. The lots shall be inspected in accordance with 4.5.2.1.3. The unit of product for the examinations in 4.5.2.1.1 and 4.5.2.1.2 shall be one finished backing plate.

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4.5.2.1: Examination of backing plate for defects in color, finish, design, construction, and workmanship. Defects designated by an asterisk (*) shall be classified as major when seriously affecting appearance or serviceability and minor when affecting appearance or serviceability but not seriously.

Table I Classification of defects.

Examine	Defect	Classification Major (*) Minor		
Color and finish	Does not match standard sample or appropriate SAE-AMS-STD-595……	X	-	-
	Not smooth……	X	-	-
	Not uniform shade……	X	-	-
	Beveled section does not have a high uniform luster……	X	-	-
	Discoloration such burn, spot, stain, streak or speck clearly noticeable……	X	-	-
	Any imbedde foreign matter……	-	*	-
	Any crack, break, blister, lump, pit, carter, or other imperfection clearly noticeable……	X	-	-
	Any scratch on beveled surface……	-	*	-
Workmanship	Any chip, nick, dent, dig, gouge, sink mark, short, flash, fin, roughness, or burr clearly noticeable……	-	-	X
	Any warp, twist, unevenness, or distortion……	-	-	X
	Any hole not formed as indicated on drawing……	-	*	-
	Knockout pin or breakoff mark located in areas visible when in actual use……	X	-	-
	Not trimmed within1/64 inch of adjacent areas……	-	*	-
	Edge of hole not clean or smooth, i.e., any burr or flush fin.	-	*	-
	Breakoff mark not neatly accomplished, i.e., sharp or rough projection which is clearly noticeable or could possibly abrade cloth material on which worn……	X	-	-

4.5.2.1.2: Examination of the backing plate for defects in dimensions. Any 4 dimension which is not within the specified tolerance shall be classified as a defect. The sample shall be one completely finished backing plate from each mold cavity.

4.5.2.1.3: Inspection levels and acceptable quality levels (AQL's). The inspection levels and the acceptable quality levels (AQL's) expressed in defects per hundred units shall be as follows:

	Inspection Level	AQL's Major	Minor
For defects applicable to 4.5.2.1.1	II	(one class)	4.0
For defects applicable to 4.5.2.1.2	S-1	(one class)	4.0

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4.5.3 Examination for count of backing plates for intermediate containers. Backing plates packaged for shipment shall be examined to determine conformance with package markings and specified quantity. The sample unit for this examination shall be one box (intermediate container). The lot size shall be the number of intermediate containers. Any box containing less than the specified marked quantity of backing plates shall be classified as a defect. The inspection level shall be S-2 and the AQL shall be 4.0 defects per sample *unit.*

4.5.3 : Examination of Reparation for delivery requirements. . An examination shall be made to determine that packaging, packing, and marking requirements of Section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container. fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item. inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

EXAMINE	**DEFECT**
Making (exterior and interior)	Omitted, incorrect, illegible, or improper size, location, Sequence or method of application.
Materials	Any component missing. Any component damaged affecting serviceability.
Workmanship	Inadequate application of component, such as incomplete closure of container flaps, loose strapping or taping or taping, inadequate strapping. Bulging or distortion of containers.
Content	Number of intermediate packages is more or less than Required.
Weight	Net weight exceeds requirements.

4.6 : End item testing. Testing of the completely fabricated backing plates 4 performed in accordance with Table I for the characteristics shown therein. The sample unit shall be three completely fabricated One determination per sample unit shall be performed requirements are applicable to the individual unit. S-1 and AQL shall be 4.0 defects per hundred units.

4.7 : Tests.

4.7.1 : Stability at high temperatures and humidity. The backing plate shall be tested in accordance with ASTM-D-756 and shall then be examined to determine compliance with 3.4.1.

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4.7.2 : Test for water absorption. The backing plate shall be tested *in* accordance with ASTM-D-570 and shall then be examined to determine compliance with 3.4.2. Dimensional changes in the backing plate shall not be considered and report.

4.8 : Packaging Inspection. The preservation, marking and marking shall be inspected to verify conformance to the requirements of Section 5.

5: PACKAGING

5.1: Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6: NOTES.

6.1: Intended use. Intended use. The Type I and Type II backing plates covered by this specification are intended for the use of enlisted infantry personnel to mount their cap and branch of service insignia.

6.2: Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, style, class, subclass (when applicable) and size of the button required (See 2.1).
(c) When standard samples will not be furnished (See 3.1).
(d) Whether standard sample is to be furnished (See 3.2).
(e) Drawing (See 2.1.2 and Figure 1).
(f) Selection of applicable levels of packing (See 5.1).

6.3 : First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first production sample or a standard production item from the contractor’s current inventory (see 3.1), and the number of items to be tested as specified in 4.4. The first article should consist of one completed backing plate. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.4: For access to standard samples, address the contracting activity issuing the invitation to bid (see 3.2).

6.5: Subject term (key word) listing.

Plastic disk

Disk

6.6: Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

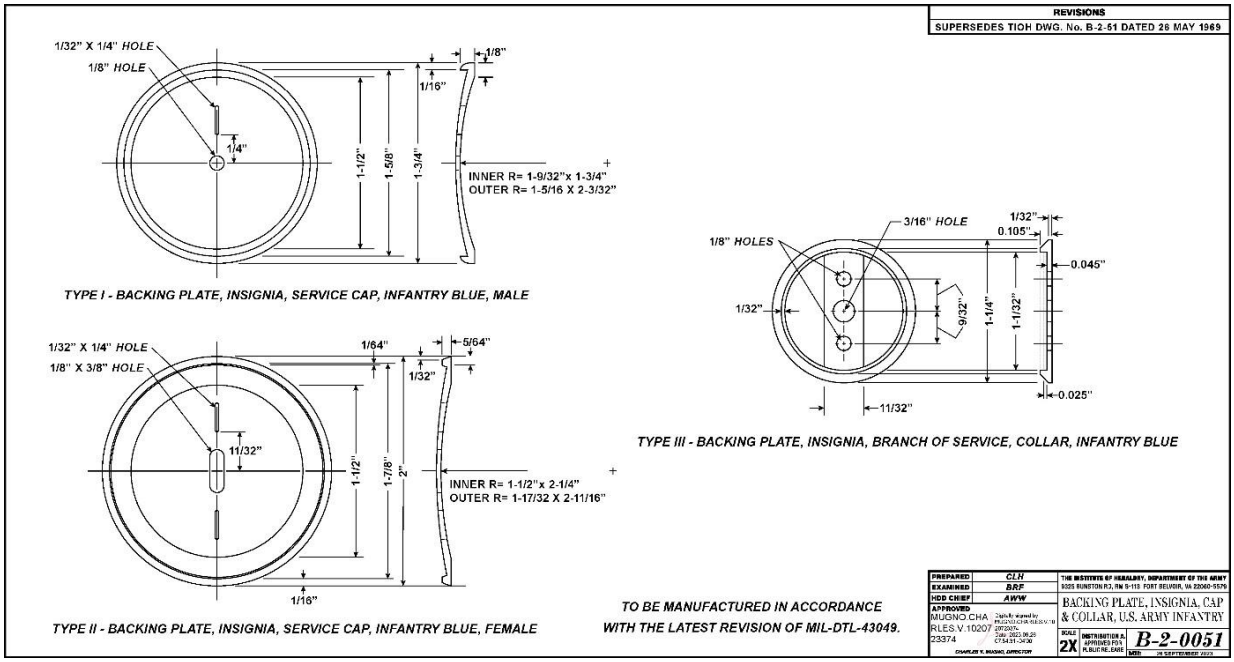


FIGURE 1 Backing Plate, Insignia, Cap & Collar, U.S. Army Infantry

Custodian:
Army - IH

Preparing activity:
Army - IH

Review activities:
DLA - CT

(Project No. 8455-2024-022)

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